Stochastic Processes Semester 1

Tutorial 9 – Solutions

1. (a) The stationary distribution within class $C_1 = \{5, 6\}$ is

$$(0, 0, 0, 0, 3/5, 2/5)$$

and within the absorbing class $C_2 = \{4\}$ is

$$(0, 0, 0, 1, 0, 0).$$

So the general family of stationary distribution is given by

$$(0, 0, 0, \gamma_1, 3\gamma_2/5, 2\gamma_2/5)$$

where $\gamma_1, \gamma_2 \geq 0$ and $\gamma_1 + \gamma_2 = 1$.

For the embedded Markov chain, the stationary distribution within class $C_1 = \{5, 6\}$ is $(0, 0, 0, 0, 1/2, 1/2)$ and within the absorbing class $C_2 = \{4\}$ is
$(0, 0, 0, 1, 0, 0)$.

So the stationary distributions within class C_1 differ in discrete and in continuous time. The reason for the difference is: $q_{55} \neq q_{66}$.

(b) For a Markov process that starts from state $X(0) = j$, the limiting distribution is

$$\left(0, 0, 0, \alpha_j^{C_2}, \frac{3\alpha_j^{C_1}}{5}, \frac{2\alpha_j^{C_1}}{5}\right).$$

(i) In Tutorial 8, we found $\alpha_1^{C_1} = \frac{16}{35}$ and $\alpha_1^{C_2} = \frac{19}{35}$. It follows that the limiting distribution starting from state $X(0) = 1$ is

$$\left(0, 0, 0, \frac{19}{35}, \frac{48}{105}, \frac{32}{105}\right).$$

(ii) In Tutorial 8, we found $\alpha_2^{C_1} = \frac{2}{35}$ and $\alpha_2^{C_2} = \frac{33}{35}$. It follows that the limiting distribution starting from state $X(0) = 2$ is

$$\left(0, 0, 0, \frac{33}{35}, \frac{2}{35}, \frac{4}{105}\right).$$

2. (a) The jumping chain is:

$$P^* = \begin{pmatrix} 0 & 1/2 & 0 & 1/2 & 0 & 0 \\ 1/4 & 0 & 1/4 & 1/4 & 0 & 1/4 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

So there is one open (transient) class $\{1, 2\}$ and 2 closed (recurrent) classes $C_1 = \{3, 5\}$ and $C_2 = \{4, 6\}$. Clearly

$$\alpha_3^{(C_1)} = \alpha_5^{(C_1)} = \alpha_4^{(C_2)} = \alpha_6^{(C_2)} = 1$$

and

$$\alpha_3^{(C_2)} = \alpha_5^{(C_2)} = \alpha_4^{(C_1)} = \alpha_6^{(C_1)} = 0.$$

Further

$$\alpha_1^{(C_1)} = 1/2\alpha_2^{(C_1)}$$

and

$$\alpha_2^{(C_1)} = 1/4\alpha_1^{(C_1)} + 1/4.$$

So $\alpha_1^{(C_1)} = 1/7$ and $\alpha_2^{(C_1)} = 2/7$, and thus $\alpha_1^{(C_2)} = 6/7$ and $\alpha_2^{(C_2)} = 5/7$.

(b) One computes the stationary distributions for each closed class using both $\boldsymbol{\pi}Q = 0$ and $\boldsymbol{\pi}P^* = \boldsymbol{\pi}$. One sees that for both situations one has $\pi_3 = \pi_5 = 1/2$ and $\pi_4 = \pi_6 = 1/2$ and so stationary distributions for $X(t)$ are of the form

$$(0, 0, \gamma_1/2, \gamma_2/2, \gamma_1/2, \gamma_2/2)$$

where $\gamma_1, \gamma_2 \geq 0$ and $\gamma_1 + \gamma_2 = 1$, while stationary distributions for X_n^* are also of the form

$$(0, 0, \gamma_1/2, \gamma_2/2, \gamma_1/2, \gamma_2/2),$$

i.e. they are equal. This is because $q_{33} = q_{55}$ and $q_{44} = q_{66}$.

(c) We use the absorption probabilities from part (a) and substitute them into γ_1 and γ_2 in (b):

i. $(0, 0, 1/7, 5/14, 1/7, 5/14)$

ii. $(0, 0, 1/2, 0, 1/2, 0)$

3. (a) We may say that there is a success if there is no offsprings. Then the time to reach 0 is the time to get the first success – it has a geometric distribution with parameter $p = 1/2$. Therefore, its mean is $1/p = 2$.
(b) The first time τ_1 when $X_{\tau_1} < 2$ has a geometric distribution with parameter $p = 3/4$ (since the probability to stay at the level 2 is equal to the probability that each individual produces one offspring, which is $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$). At time τ_1, there is a jump to the level 0 with conditional probability $\frac{1/4}{3/4} = \frac{1}{3}$ (this is the probability that both individuals do not produce offsprings given that at least one does not), and the process stops. Otherwise (with probability 2/3) there is a jump to the level 1 and then the process reach state 0 after a further geometric random variable τ_2 with parameter 1/2. So the total time T is τ_1 with probability 1/3 and $\tau_1 + \tau_2$ with probability 2/3. Therefore,

$$\mathbf{E}T = \mathbf{E}\tau_1 + \frac{2}{3}\mathbf{E}\tau_2 = \frac{4}{3} + \frac{2}{3} \cdot 2 = \frac{8}{3}.$$

4. (a) This is a geometric random variable τ with parameter $p = p_2 = 2/3$, so $\mathbf{E}\tau = 3/2$.
(b) The process stays at level 2 if each individual produces exactly one offspring, i.e. with probability $p_1^2 = 1/9$. So the probability of 'success' is $p = 1 - 1/9 = 8/9$ and the expected time to exceed the level 3 is 9/8.
(c) The first time τ_1 when $X_{\tau_1} > 2$ has a geometric distribution with parameter $p = 8/9$ (since the probability to stay at the level 2 is equal to the probability that each individual produces one offspring, which is $\frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$). At time τ_1, there is a jump to the level 4 with conditional probability $\frac{4/9}{8/9} = \frac{1}{2}$ (this is the probability that each individual produces 2 offsprings given that at least one does), and the process stops. Otherwise (with probability 1/2) there is a jump to the level 3 and then the process reach set $\{4, 5, 6, \ldots\}$ after a further geometric random variable τ_2 with parameter $p = 1 - (1/3)^3 = 26/27$. So the total time T is τ_1 with probability 1/2 and $\tau_1 + \tau_2$ with probability 1/2. Therefore,

$$\mathbf{E}T = \mathbf{E}\tau_1 + \frac{1}{2}\mathbf{E}\tau_2 = \frac{9}{8} + \frac{1}{2} \cdot \frac{27}{26} = \frac{171}{104}.$$

5. (a) $\mu = \mathbf{E}\xi = 1/4 + 2(1/4) = 3/4 < 1$ and so $\varepsilon = 1$
(b) $\mu = 1/3 + 2(1/3) = 1$ and so $\varepsilon = 1$
(c) Here the probability is ε^3 which is again 1, by (b)
(d) Here $\mu = 1/4 + 2(1/2) = 5/4 > 1$, so the extinction probability ε is less than 1, and is the minimal solution of the equation $\varepsilon = \varphi(\varepsilon)$ where $\varphi(z) = \mathbf{E}z^{\xi}$ is the probability generating function of the number of offsprings of a typical individual. In our case $\varphi(z) = \frac{1}{4} + \frac{z}{4} + \frac{z^2}{2}$, so we have to find the minimal root of the equation

$$\varepsilon = 1/4 + \varepsilon/4 + \varepsilon^2/2$$

which is

$$\varepsilon = \frac{3/2 - \sqrt{9/4 - 2}}{2} = \frac{1}{2}.$$

(e) Here the probability of extinction is $\varepsilon^4 = 1/16$ where $\varepsilon = 1/2$ is from (d).
(f) Here $\mu = 1/5 + 2(3/5) > 1$ and so ε is the minimal root of the equation

$$\varepsilon = 1/5 + \varepsilon/5 + 3\varepsilon^2/5$$

and so $\varepsilon = 1/3$.
(g) Here the probability is ε^5 which is $(1/3)^5 = 1/243$, using (f)